

EC4010 Digital Design

Assignment-1

Q1: Design a 2-to-1 multiplexer using Verilog. The multiplexer should have two data inputs (A , B), one select input (S), and one output (Y).

- If $S = 0$, $Y = A$
- If $S = 1$, $Y = B$

Q2: Design a 7-segment display decoder using Verilog that takes a 4-bit input ($A[3:0]$) and drives the appropriate segments to display the corresponding hexadecimal digit (0-9, A-F). The output should be in the form of a 7-bit signal (segments $[6:0]$), where each bit corresponds to a segment on the 7-segment display.

Q3: Design an 8-bit Arithmetic Logic Unit (ALU) using Verilog that performs the following operations based on a 3-bit operation code ($Op[2:0]$):

- $Op = 000$: ADD
- $Op = 001$: SUBTRACT
- $Op = 010$: AND
- $Op = 011$: OR
- $Op = 100$: XOR
- $Op = 101$: NAND
- $Op = 110$: NOR
- $Op = 111$: NOT

The ALU should have two 8-bit inputs ($A[7:0]$, $B[7:0]$) and an 8-bit output ($Result[7:0]$).